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The Effects of Pulse Width on Fibre-optic Distributed Vibration Sensing Data

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SUMMARY

Systems for distributed vibration sensing (DVS), also known as distributed acoustic sensing (DAS), use a fibre-optic cable as the sensor, rather than the discrete point sensors used in conventional systems. Theoretically, they measure strain, which is the change in the length of the section of fibre divided by the gauge length. In reality, they measure the change in phase of the backscattered light which is directly proportional to strain. However, the phase is not measured at a single position along the fibre; instead, the value is a result of the application of a Gaussian window.

The results included here indicate that the laser pulse width used during DVS acquisition can have a significant effect on the quality of the data acquired. As shown previously, the gauge length must be carefully selected due to its strong frequency response, but due to the Gaussian nature of the pulses, their associated frequency response is more muted. The point at which the effect of the pulse width becomes larger than the gauge length is half the gauge length, allowing considerable scope for it to be increased, and thus the SNR of the data improved.

Introduction

Systems for distributed vibration sensing (DVS), also known as distributed acoustic sensing (DAS), use a fibre-optic cable as the sensor, rather than the discrete point sensors used in conventional systems (Hartog et al., 2014). Theoretically, they measure strain, which is the change in the length of the section of fibre ΔL divided by the length of the section of fibre over which the measurement is being made $L_2 - L_1$, the gauge length. In reality, DVS systems measure the change in phase $\Delta\phi$, which is directly proportional to ΔL . ΔL is equal to the integral of the infinitesimal strain ϵ along the section of fibre. Displacement s is the spatial integral of the strain and thus ΔL is the difference between displacement measurements at either end of the fibre section. Therefore, the phase at each of these positions is proportional to displacement; mathematically

$$\Delta\phi \propto \Delta L = \int_{x=L_1}^{L_2} \epsilon \, dx = s_2 - s_1 \propto \phi_2 - \phi_1 \quad (1)$$

Figure 1a shows a strain (Ricker) wavelet plotted against depth. The gauge length is shown as a green box; the resulting displacement measured at both ends of the gauge length is shown in Figure 1b and the strain and ΔL in Figure 1c. The gauge length has the effect of smoothing the wavelet because it is, in effect, a moving average filter applied in the spatial domain (Dean et al., 2015a). However, the phase is not measured at a single position along the fibre; instead, the value is a result of the application of a Gaussian window, as shown in Figure 1d, representing the width of the laser pulse used to interrogate the fibre. This results in further smoothing of the wavelet (Figure 1f). In this paper we look at the effects of the choice of the pulse width on the quality of DVS data.

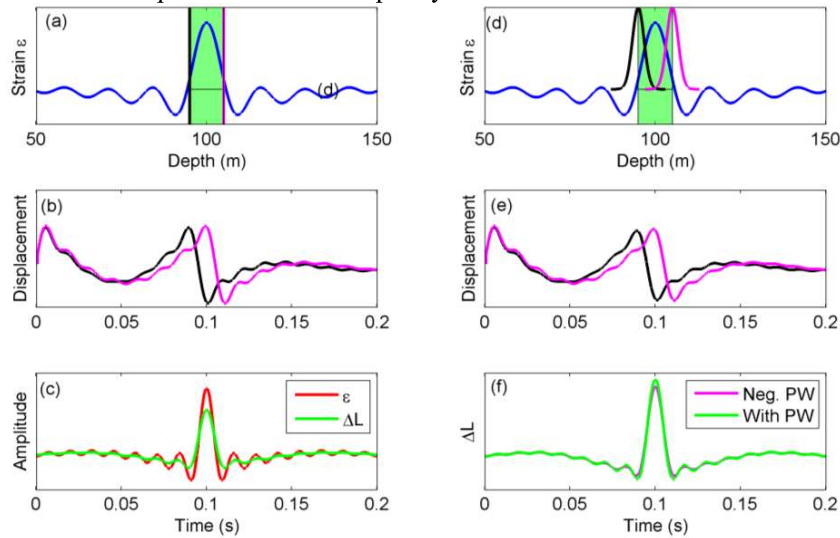


Figure 1 Simple example showing a Klauder wavelet (8 to 80 Hz) travelling at 1,000 m/s. The green box in (a) and (d) indicates the extent of the gauge length (10 m). The colours of the displacement plots shown in (b) and (e) correspond to the windows shown in (a) and (d), respectively. The strain measured at the position shown by the red line in (a) is shown with the same colour in (c) along with the value of ΔL . (f) shows the resulting ΔL neglecting and including the pulse width, the extent of the latter being shown in (d).

Effects on Resolution

The use of a Gaussian window along the fibre length results in a filter being applied to the data in the wavenumber domain. The equation for a Gaussian window in the time domain is

$$A_t(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \quad (2)$$

where σ is the standard deviation which is related to the half-width h by $\sigma = h/2.3548$. The frequency domain response is

$$A_f(f) = e^{-\frac{f^2}{2\sigma_f^2}} \quad (3)$$

where f is the frequency, and σ_f is the standard deviation in the frequency domain, which is given by

$$\sigma_f = \frac{1}{2\pi\sigma}. \quad (4)$$

Of course, the gauge length L also has a wavenumber response (Dean et al., 2015b)

$$A_{gl}(k) = \frac{\sin(\pi k L)}{\pi k}. \quad (5)$$

Figure 2 shows Klauder wavelets generated using pulse widths of between 2 and 10 m. Increasing the pulse width has little effect on the width of the wavelet because the frequency domain response is mostly governed by the effects of the gauge length (Figure 2c).

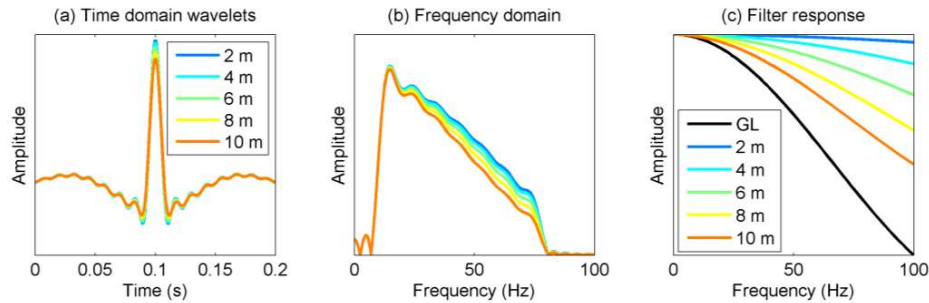


Figure 2 Examples of Klauder wavelets (8 to 80 Hz) travelling at 2,000 m/s for a gauge length of 20 m. Although effects from the pulse width are evident, the spectrum of the resulting wavelet is principally affected by the gauge length.

As shown in Figure 3 the response of the gauge length at low wavenumbers can be approximated by a Gaussian; thus, to determine the relative effects, we can compare the 3-dB down points resulting from the two parameters. The 3 dB down point of the pulse width equivalent Gaussian filter is given by

$$f_{3dB} = \sigma_f \sqrt{2 \ln \sqrt{2}}, \quad (6)$$

The 3-dB down point for the gauge-length moving-average response is best solved numerically. Figure 3 shows the relative responses for the typical acquisition parameters of a pulse with a half-width of 8 m and a gauge length of 30 m. The 3-dB down point of the former occurs at 0.038 cycles/m but the latter occurs at only 0.015 cycles/m.

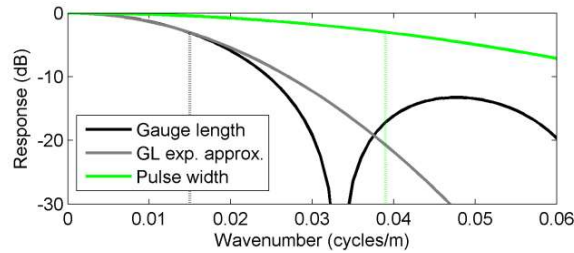


Figure 3 The wavenumber response resulting from a pulse half-width of 8 m and a gauge length of 30 m.

Figure 4 shows the 3-dB down point calculated for pulse widths between 1 and 25 m (a) and gauge lengths of between 10 and 100 m (b). Figure 4c summarises the pairs of values at which the gauge length (blue section) or the pulse width (green section) has the greatest effect. The gradient of the dividing line is ~ 0.56 , i.e., as long as the pulse width is not more than half the gauge length, the latter will have the greatest effect.

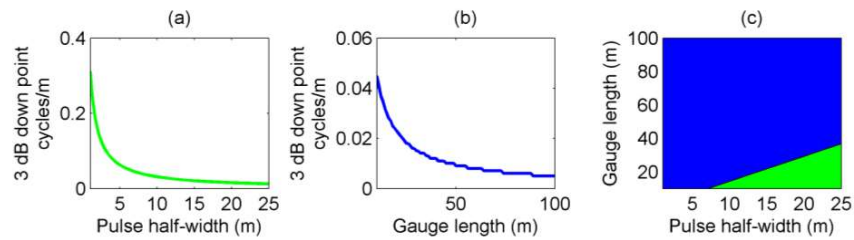


Figure 4 The 3-dB down point for different (a) pulse widths and (b) gauge lengths. (c) shows which parameter has the greatest effect for different combinations of gauge length and pulse width; blue is the gauge length and green the pulse width.

Effects on SNR and Resolution

Figure 5 shows synthetic data generated using an 8 to 80 Hz Klauder wavelet calculated using a gauge length of 20 m for three different pulse widths. For the 1 m pulse width the data are extremely noisy, but improve considerably when the pulse width is increased. Figure 6 shows summary plots of (a) signal and noise strengths, (b) signal-noise ratio (SNR), and (c) the period of the wavelet. The integral of a Gaussian window is approximately equal to that of a rectangular window with the same half-width. Assuming random noise the decrease in noise level should therefore be proportional to the square root of the pulse width. Due to the shape of the response curve (e.g., Figure 2), the effect on the signal is minimal, both in terms of strength and resolution, resulting in the SNR also varying with the square root of the pulse width.

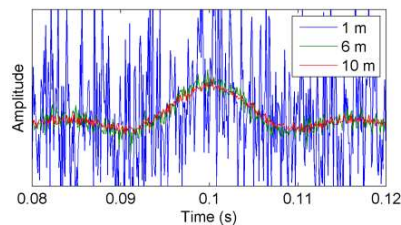


Figure 5 Three synthetic Klauder wavelets (8 to 80 Hz) calculated using a gauge length of 20 m, a velocity of 2,000 m/s, and pulse widths of 1, 6, and 10 m.

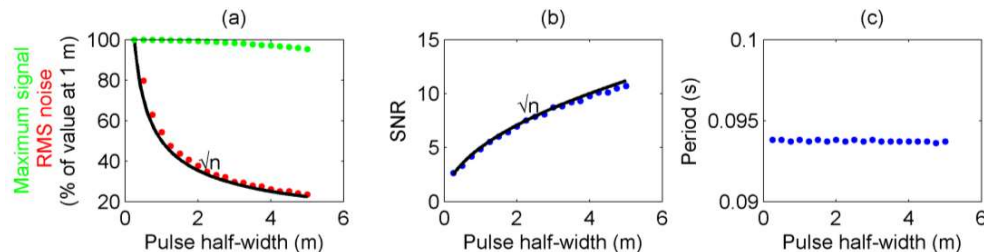


Figure 6 The variation in (a) signal and noise strength, (b) SNR, and (c) period (measured between zero crossings) for different pulse widths at a gauge length of 10 m.

Laboratory Data

The effects of pulse width were tested in the laboratory by recording data using a section of fibre within which a short-length fibre-stretcher had been placed that was driven at a frequency of 100 Hz. The data were all processed with a gauge length of 6.1 m, which is equivalent to 60 ns. Figure 7 shows two example traces, one with the fibre-stretcher and one without for two pulse half-widths, 11 ns and 75 ns. The effect on the SNR is significant, with the larger pulse width having a value 23 dB higher.

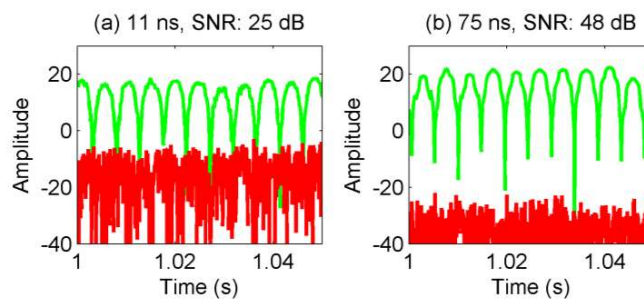


Figure 7 An example of the DVS data recorded in the laboratory for two different pulse widths. The green traces are the section of the fibre affected by the fibre-stretcher, the red traces are noise. The fibre stretcher was being driven by a sin wave, but the absolute values are shown here.

Figure 8 shows a summary of the SNR of the laboratory data; the results follow the square-root relationship as predicted from the modelled results apart from the shortest pulse width, which is probably too short to enable accurate phase estimation.

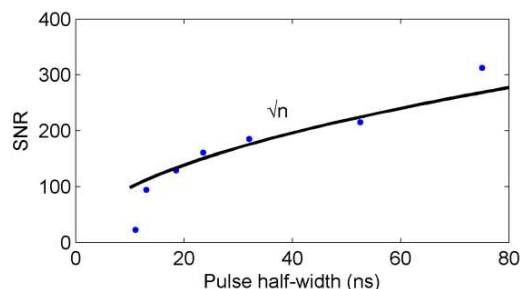


Figure 8 Summary plot of the laboratory data acquired with different pulse widths.

Discussion and Conclusions

The results shown here indicate that the laser pulse width used during DVS acquisition can have a significant effect on the quality of the data acquired. As detailed in Dean et al. (2015) the gauge length must be carefully selected due to its strong frequency response, but due to the Gaussian nature of the pulses, their associated frequency response is more muted (Figure 3). The point at which the effect of the pulse width becomes larger than the gauge length is half of the gauge length, allowing considerable scope for it to be increased, and thus the SNR of the data improved.

The ease of changing the pulse duration will depend on system design. Most DVS/DAS systems (Hartog et al., 2013) have pulses defined by electronically controlled modulators which, in principle, can be reprogrammed. However, other aspects of the system, such as the design of any optical amplifiers used to boost the pulse energy, may also need to be altered.

Acknowledgements

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